Daniel Brown

Senior Software Engineer focused on production AI systems (LLM assistants, agentic workflows, and RAG), alongside performant, scalable web and mobile experiences.

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Experience

Senior Software Engineer - Future Secure AI
Sept 2025 - Present

- Design and build production LLM-powered assistants and multi-agent workflows (LangChain, LangGraph, n8n, Flowise) for task and process automation.
- Implement RAG pipelines, tool/function calling, and agent planning so models can retrieve grounded context and take reliable actions.
- Apply advanced RAG techniques, LLM context management, and safety guardrails for controllable, production-ready model behaviour.
- Integrate and operate advanced AI models (including document AI / AWS Textract) with a focus on scalable, reliable delivery.
- Migrate and modernise legacy systems toward AI-enabled platforms and intelligent automation.

Senior Software Engineer - Tiny Technologies
Aug 2023 - Aug 2024

- Maintenance and improvement of an established, large-scale codebase.
- Extended application of TypeScript and web technologies to consistently meet client needs.
- CI/CD and continuous integration practices; automated tests (Playwright) for reliability and quality.
- Led code reviews and mentored junior developers; supported team performance and code health.

Tutor for ITAP - University of Wollongong
Mar 2020 - June 2022

- Experience with teaching indigenous University students.
- Required to set up a meeting schedule with students.
- Required excellent communication skills.
- Required to differ teaching to suit different abilities and needs.

Senior Software Engineer - FrogCo
Aug 2024 - July 2025

- Development of mobile apps using React Native and React Native Reanimated with gestures.
- Integrating and optimising video streaming in mobile apps.
- Maintenance and deployment of mobile apps for iOS and Android.
- Developing with a large number of users and network requests in mind.
- Usage of mobile specialised continuous integration utilities and services.
- Development of Server Side Rendered NextJS web apps.
- Containerisation of web app stacks using Docker and docker compose configurations.

Consultant - Auxilis Feb 2024 - Sept 2024

- Strategic direction and technical advice for existing project operations and roadmaps.
- Guiding design pattern decisions to optimise efficiency, readability and maintainability.
- High-level application of Python, TypeScript & Node; mentorship and cross-functional alignment with engineering and stakeholders.

Team Lead - Auxilis June 2022 - Aug 2023
Senior Software Engineer Mar 2020 - June 2022
Software Engineer June 2019 - Mar 2020
Trainee Software Engineer June 2018 - June 2019

- Advanced use of Javascript, Typescript, Python & NodeJS to deliver web-apps and serverless backends.
- Experience working in a startup environment in Agile delivery; regular communication with internal executives and external stakeholders.
- Utilised Typescript & Node to deliver scalable backend architectures & microservices.
- Led and mentored developers; tutoring, career development and cross-functional collaboration with Product, Engineering and Operations.
- Presented complex technical projects to stakeholders and remote teams.
- Designed and operated production AI-driven document processing (IDP) solutions for complex OCR and extraction at scale.
- Prisma backend and GraphQL; end-to-end tests (Playwright) and quality practices for production reliability.

Skills

Strongest languages
9 years in JavaScript (ES6+), 8 years in TypeScript, 6 years in Python, 4 years in Golang

Other Known Languages
Nix, Java, C++, Lua, Rust, Nim

Libraries & Frameworks
React, Mocha, React Native (Expo), Angular, Playwright, Tailwind CSS, Vitest, Jest, Pocketbase, Svelte, SvelteKit, Vue, SST, Capacitor (Ionic), React Native Reanimated, Tanstack Query, Prisma, GraphQL, Next.js, Express

Tools & Platforms
Amazon Web Services, AWS Lambda, AWS DynamoDB, AWS S3, Node.js, Microservices, Serverless, Google Cloud Vision OCR, PostgreSQL, AWS Textract, Firebase, NPM publishing, Linux, Windows Subsystem for Linux (WSL), Debian, CI/CD, Artificial Intelligence (AI), PNPM, NixOS, Bun, Docker, Optical Character Recognition (OCR)

Education

University of Wollongong
Bachelor of Computer Science - 2020

AWS
AWS Partner: Accreditation (Technical) - 2019

Open University
Understanding Autism - 2024

Selected Projects

ai-quiz
Research any subject into a local fact bank, then play agent-authored quizzes in chat or a local web UI. A researcher agent gathers facts through a CLI; a quiz host samples that bank and writes multiple-choice, yes/no, and freeform rounds with an OpenAI-compatible LLM or the Cursor Agent SDK.

Polyphasic Sleep Manager
An iOS app written in React-Native to help manage a polyphasic sleep schedule with your own personal schedule. Integrates with your phone’s calendar to see where you can fit in naps during the day and how to best fit in your sleep schedule.

FrogBox Play - FrogCo
A mobile app for watching cricket clips and highlights. Videos are collected from livestreams across the UK and Australia, then clips are extracted using AI.

Lens of Brown Photography
A website I wrote from scratch with Svelte-kit for my photography business. It's a static website hosted on Github pages, with a score of 100 on Google lighthouse.

View all my projects

Beyond work

Volunteering
Active WIRES volunteer, certified for Rescue and Immediate Care (native wildlife) and Identification and Treatment of wombat mange. Passionate about helping native wildlife, providing education to the community and assisting with wider population conservation projects.

Interests
Outside of coding, I enjoy photography, playing video games, snowboarding, rock climbing and spending time with my wonderful wife and dog.